



Figure 10: **Iterative vs Single-Step Mixing.** Solid lines represent the experimental results (5 runs), while dashed lines correspond to the theoretical predictions of Corollary 3. The iterative mixing is repeated 5 times, with $p_1 = p_2 = 0.5$. “Clean” refers to the scaling when using solely the $n_1 = p_1 n$ real data in the dataset.

We now explore some important consequences of Corollary 3.

Iterative Mixing Recovers Scaling Laws but Might Not be Feasible in Practice. If the practitioner can curate a sufficient amount of data from the original distribution, the training dataset will include a non-vanishing proportion of real data, ensuring that p_1 remains bounded away from zero. By comparing $\bar{E}$ with p_2^{2t} , we observe that iterative mixing over t iterations, where t is of the order of $\log(n/d)$, results in a scaling law proportional to $\bar{E}$, as empirically confirmed in Figure 10. However, this comes at the cost of significant bootstrapping, a large volume of real data, and the need to clearly distinguish between real and synthetic data across iterations—conditions that are all too computationally expensive and challenging to implement in practice.

Iterative Mixing Relies Mostly on Real Data. In Figure 10, we compare the scaling of iterative mixing with the scaling obtained using only the $p_1 n$ real data portion from the same training set (“Clean”). While the scaling rate remains consistent, iterative mixing consistently underperforms compared to using real data alone. This suggests that iterative mixing may primarily neutralize the synthetic data and heavily rely on the real data to recover the scaling. Even when the original synthetic data is of high quality (i.e., when c_0 is small, rightmost plot of Figure 10), the iterative method fails to effectively leverage the synthetic data, resulting in worse performance than single mixing. Thus, although iterative mixing recovers the same scaling rate, the model still collapses to some degree, and no significant performance improvement is observed.

Iterative Mixing with Little Real Data is Bad. If we consider the setting where we only have limited real data or where there is faster accumulation of synthetic data, which corresponds to $p_2 \rightarrow 1$ (the real data in the training set is diminishing), then it holds that for any $t \geq 1$, $E_{test}(\hat{w}^{(t)}) \simeq c_0^2 + t\bar{E}$. This is an increasing function of t , meaning that there is still catastrophic model collapse.

6 Discussion

Our work systematically characterizes the effects of training models on mixtures of real and synthetic data, showing that model collapse is a robust phenomenon that persists even with small fractions of synthetic data, in the asymptotic regime. By introducing new mathematical tools, we extend prior work to analyze more complex mixing settings and models (random projections), broadening the scope of theoretically tractable problems. Experiments confirm our theoretical predictions across large language models (LLMs) and also fully-trained feed-forward neural networks.

Going beyond the prevalent “neural scaling laws” paradigm (Kaplan et al., 2020; Hoffmann et al., 2022) which is at the basis of the current trend in training LLMs, this study emphasizes the importance of preserving and labeling real data, either by curating it or avoiding unintended synthetic data in training, reflecting a shift as AI-generated data becomes prevalent. Our work also delineates the impact of model size on the model collapse profile. Future work will explore the effect of other model design choices like activation functions, depth, and optimization hyper-parameters like learning rate and momentum. To this end, we can leverage “Gaussian equivalents” (Goldt et al., 2022) to extend our theory to wide, fully-trained networks in the neural tangent kernel (Jacot et al., 2018) and lazy (Chizat et al., 2019) regimes, using operator-valued free probability theory (Mingo & Speicher, 2017), like we have done in our analysis.

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